

Computer Networks

A practical study reference with foundational concepts, examples, and common review questions.

1. Layers and Addressing

The OSI model describes physical, data-link, network, transport, session, presentation, and application layers. The Internet suite is commonly described as link, internet, transport, and application. Encapsulation adds headers as data moves through layers.

Ethernet and Wi-Fi carry local frames; IP routes packets. IPv4 addresses are 32 bits; IPv6 addresses are 128 bits. CIDR prefixes such as /24 identify network bits. Routers forward between networks; a default gateway handles unmatched routes.

2. Transport and Services

TCP provides reliable, ordered byte streams with retransmission, flow control, and congestion control; it does not preserve application message boundaries. UDP sends datagrams without built-in delivery or ordering guarantees.

DNS resolves names into records such as A and AAAA and may cache answers according to TTL. DHCP can assign network configuration. ARP maps IPv4 local-network addresses to link-layer addresses; IPv6 uses Neighbor Discovery.

3. Web Security and Troubleshooting

HTTP has methods, status codes, headers, and optional bodies. TLS authenticates a server and encrypts communication when correctly configured; HTTPS is HTTP over TLS. A port identifies a transport endpoint. Firewalls filter traffic but do not replace secure applications.

Troubleshoot from local interface and link, to IP and route, DNS, transport connectivity, TLS, then application response and logs. Expose only required services and never ignore certificate warnings without understanding the cause.

4. Common Questions

TCP is a reliable ordered byte stream; UDP is datagram-based without built-in reliability. DNS maps names to records. A subnet prefix specifies the network portion of an IP address. NAT translates addresses at network boundaries but is not itself a firewall or security guarantee.